



Review

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*Corresponding author : Nana Chung

Department of Physical Education, Sangji University, Wonju, Republic of Korea

Tel: +82-33-730-0637

E-mail: cnmms44@sangji.ac.kr

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Energy availability as an integrative framework: reinterpreting the energy balance and carbohydrate–insulin models in athletes

Nana Chung*

Department of Physical Education, Sangji University, Wonju, Republic of Korea

[Purpose] This review reinterprets the Energy Balance Model (EBM) and the Carbohydrate–Insulin Model (CIM) within athletic contexts and evaluates their explanatory relevance for low energy availability (LEA) and relative energy deficiency in sport (RED-S).

[Methods] We conducted a structured conceptual review using a narrative synthesis approach. Literature searches were performed through January 2026 in PubMed, Web of Science, and SPORTDiscus, using terms related to EA, LEA, RED-S, EBM, CIM, and athlete metabolic, endocrine, and performance responses.

[Results] Reductions in EA consistently induce endocrine, metabolic, and performance disturbances, independent of body mass changes. EBM primarily accounts for long-term regulatory adaptations associated with cumulative energetic strain, whereas CIM provides mechanistic insights into short-term alterations in substrate partitioning and hormonal signaling. Across sports disciplines, LEA has emerged as a coordinated multisystem physiological adaptation rather than an isolated consequence of reduced energy intake.

[Conclusion] LEA and RED-S are most coherently interpreted within an EA-centered framework that integrates long-term energy regulation and short-term metabolic dynamics. This integrative perspective provides a unified physiological basis for understanding the metabolic adaptation, performance regulation, and health outcomes of athletes.

[Keywords] energy availability, low energy availability, relative energy deficiency in sports, energy balance model, carbohydrate–insulin model, metabolic adaptation, endocrine responses, athletic performance

INTRODUCTION

The Energy Balance Model (EBM) and Carbohydrate–Insulin Model (CIM) have long served as central theoretical frameworks for explaining changes in body weight and metabolic adaptation. EBM is based on thermodynamic principles, which propose that alterations in body weight and body composition arise from the cumulative imbalance between energy intake and expenditure over time¹. In contrast, CIM emphasizes the role of carbohydrate intake and insulin signaling in regulating substrate partitioning, energy storage, and metabolic state, thereby influencing metabolic outcomes beyond total energy intake alone². Historically, these models have been discussed in opposition to each other, particularly in the context of obesity research and public health. More recently, however, increasing attention has been directed toward reevaluating their explanatory scopes and limitations^{2,3}.

Despite this renewed interest, the EBM–CIM debate has largely developed in the context of the general population, with limited consideration of athletic environments. Athletes are exposed to a distinct constellation of physiological stressors, including high total energy expenditure, repeated training and recovery cycles, sport-specific substrate demands, and, in many cases, intentional body weight manipulation. Under such conditions, changes in body weight do not always accurately reflect the underlying energy status or physiological adaptations. Performance decrements, endocrine disturbances, and health risks are frequently observed even in the absence of marked weight loss, highlighting the important limitations in applying traditional weight-centric interpretations to athletes^{4,5}.

The concepts of low energy availability (LEA) and relative energy deficiency in sports (RED-S) further underscore these limitations. LEA refers to the state in which insufficient energy remains available to support normal physiological functions after accounting for the energy cost of exercise. Importantly, LEA can induce endocrine, metabolic, and performance impairments, even when body weight remains stable⁶. RED-S extends this framework by encompassing a broad spectrum of physiological consequences, including disruptions in reproductive function, bone health, metabolic regulation, immune function, and athletic performance.

EA as the Integrative Interface Between EBM and CIM in Athletes

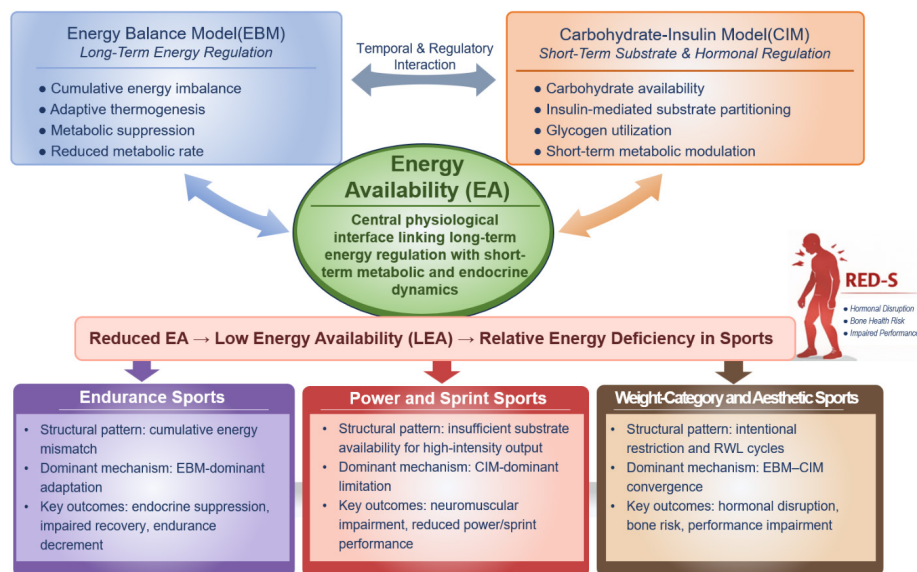


Figure 1. EA as the integrative physiological interface linking the long-term regulatory domain of EBM with the short-term metabolic and endocrine domain of CIM in athletes. Reduced EA underlies LEA, the systemic manifestations of RED-S, and their sport-specific expression across endurance, power and sprint, and weight-category/aesthetic sports.

Together, these constructs challenge the assumption that body weight change is a necessary indicator of energy deficiency, and highlight the need for models that more accurately reflect the physiological realities of trained athletes.

However, the roles of EBM and CIM in explaining LEA and RED-S in athletes remain unclear. Existing interpretations variably emphasize chronic energy deficits and long-term metabolic adaptation or focus on short-term alterations in substrate utilization and hormonal signaling, often resulting in fragmented explanations.

Although previous discussions on EBM and CIM have provided valuable insights into body-weight regulation and metabolic adaptation, their synthesis has rarely been explicitly developed for athletic populations, in which body mass is often an insensitive indicator of energetic strain. In this context, the present review does not revisit the EBM–CIM debate as a question of theoretical superiority. Rather, it proposes a three-level integrative framework: first, EBM and CIM are reinterpreted as complementary models operating at different explanatory and temporal scales; second, energy availability (EA) is positioned as the central physiological mediator linking long-term regulatory adaptation with short-term metabolic and endocrine dynamics; third, this integrative perspective is translated across sport disciplines to clarify how LEA and RED-S are expressed under different training and performance demands.

Accordingly, the purpose of this review is to reinterpret EBM and CIM within athletic contexts as complementary models linked through EA. By synthesizing evidence across sports disciplines, this review aims to clarify the sports-specific manifestations of LEA and RED-S and to provide a unified physiological framework for understanding how

long-term energy regulation and short-term metabolic dynamics interact in athletes. The conceptual model underlying this interpretation is shown in Figure 1.

To support this conceptual synthesis, a structured literature search with a narrative review approach was conducted through January 2026 in PubMed, Web of Science, and SPORTDiscus, using terms related to EA, LEA, RED-S, EBM, CIM, and athlete metabolic, endocrine, and performance responses. Experimental studies, observational studies, reviews, and consensus statements were considered when addressing athlete physiology, EA-related mechanisms, and sport-specific manifestations of LEA and RED-S. Studies that focused exclusively on general obesity treatment or non-athletic populations without a clear relevance to athletic metabolism, performance, or RED-S physiology were excluded. The PRISMA-ScR guidance was conceptually considered to improve transparency; however, full PRISMA-ScR procedures were not applied because this review was designed as a structured narrative synthesis rather than a formal scoping review.

THEORETICAL CONTEXT OF THE ENERGY BALANCE AND CARBOHYDRATE-INSULIN MODELS

EBM is based on the principle that changes in body weight arise from the difference between energy intake and expenditure. Sustained energy intake that exceeds expenditure results in weight gain, whereas persistent energy deficits lead to weight loss¹. A defining feature of EBM is that energy expenditure is dynamically adjusted in response

to changes in energy intake. Reductions in resting metabolic rate and non-exercise activity observed during weight loss reflect adaptive thermogenesis, a regulatory response aimed at conserving energy⁷. These adaptations may persist even after body weight stabilization.

Similar long-term regulatory responses are observed in athletes when high training loads do not match adequate energy intake. Athletes may experience metabolic suppression and impaired recovery despite relatively stable body weight. However, although EBM provides a robust framework for interpreting chronic adaptations, it offers limited insight into acute alterations in substrate utilization, endocrine responses, and performance capacity.

CIM shifts the explanatory focus toward hormonal regulation of substrate partitioning. CIM posits that dietary carbohydrate intake modulates insulin secretion, influencing fuel distribution and metabolic state^{8,9}. Experimental studies have reported differences in insulin response, fat oxidation, and glycogen utilization under varying dietary compositions^{2,3}. Such findings are particularly relevant in athletic contexts where performance is closely linked to substrate availability. Nevertheless, CIM does not fully account for the chronic adaptations associated with sustained energy deficits. Controlled trials have indicated that long-term metabolic outcomes remain constrained by total energy intake³.

Rather than representing competing explanations, EBM and CIM can be understood as complementary frameworks operating on distinct physiological and temporal scales. EBM primarily accounts for long-term regulatory adaptations, whereas CIM provides mechanistic insights into short-term metabolic dynamics. This distinction is especially relevant in athletes, in whom physiological disturbances and performance impairments may arise independent of measurable body mass changes. In this context, EA has emerged as a critical integrative construct.

ENERGY AVAILABILITY AND THE PHYSIOLOGICAL BASIS OF LOW ENERGY AVAILABILITY

EA is defined as the energy remaining for physiological functions after accounting for exercise energy expenditure and is expressed relative to fat-free mass (FFM)⁴. Conceptually, EA is distinct from body weight and total energy intake, as physiological responses may differ substantially under similar body weight conditions depending on the training load and exercise energy expenditure. Accordingly, EA is widely used to evaluate the energy status of athletes^{10,11}. Experimental studies have shown that reductions in EA to approximately 30 kcal·kg⁻¹ FFM·day⁻¹ or lower are associated with marked disruptions in endocrine and metabolic regulation¹²⁻¹⁶. However, this value should not be interpreted as the universal threshold. It was derived predominantly from studies in female exercising populations, particularly endurance-oriented cohorts and may not apply uniformly to male athletes, power-based disciplines, or conditions characterized by intermittent or within-day energy deficiency^{10,17,18}.

When EA is chronically or repeatedly reduced, athletes transition to a state of LEA, characterized by coordinated physiological adaptations that prioritize energy conservation. Both acute and sustained reductions in EA have been shown to induce rapid endocrine alterations independent of measurable body weight changes¹⁶, with similar responses reported across athletes¹¹. Therefore, LEA should be interpreted in relation to its duration and recurrence, as acute LEA may arise during short-term energy restriction or intensified training, whereas repeated or sustained LEA is associated with cumulative regulatory adaptations^{10,17,18}. In addition, recent work suggests that within-day energy deficiency may represent a distinct dimension of energetic stress, even when the total daily energy intake is less severely reduced¹⁸.

Endocrine regulation is highly sensitive to reductions in EA. LEA disrupts hypothalamic function by reducing gonadotropin-releasing hormone (GnRH) pulsatility and downstream luteinizing hormone (LH) and follicle-stimulating hormone (FSH) secretion^{10,16}. Thyroid hormone regulation is similarly affected, with reduced triiodothyronine (T3) concentrations and resting metabolic rates reflecting adaptive thermogenesis^{7,17,18}. Elevated cortisol concentrations and reductions in insulin-like growth factor-1 (IGF-1) further characterize this metabolic environment^{6,19,20}, while peripheral signals such as decreased leptin and insulin alter the hypothalamic sensing of energy status. Bone metabolism is also highly sensitive, as sustained reductions in EA are associated with suppressed bone formation, increased bone resorption, and reduced bone mineral density^{21,22}.

Therefore, sex-specific physiological responses to LEA should be considered. In female athletes, reduced EA is closely linked to hypothalamic suppression, reduced gonadotropin-releasing hormone pulsatility, menstrual dysfunction, estrogen deficiency, and impaired bone health^{10,16,22}. In male athletes, LEA has been associated with reduced testosterone and IGF-1 levels, a more catabolic endocrine profile, impaired muscle protein synthesis and recovery, and an altered cortisol-to-testosterone balance^{17,19,20,23}. These findings support the view that RED-S is not sex-specific in origin, although its physiological expression may differ by sex²⁴.

Collectively, these findings indicate that LEA represents a coordinated multi-system physiological adaptation rather than an isolated dysfunction, which forms the biological foundation of RED-S⁶. From an integrative perspective, EA functions as a physiological interface that links long-term energy regulation with short-term metabolic and endocrine dynamics because reductions in EA elicit coordinated responses underlying both LEA and RED-S.

SPORT-SPECIFIC MANIFESTATIONS OF LOW ENERGY AVAILABILITY AND RED-S

The physiological framework outlined above suggests that the consequences of LEA cannot be interpreted uniformly across athletes. Although reduced EA represents a common initiating condition, the manifestation of LEA and RED-S varies according to the metabolic demands of the

sport, the structure of training and recovery, and the extent to which body weight or physique is regulated as part of performance preparation.

On this basis, the present review adopted a three-category sport classification to facilitate the interpretation of sport-specific patterns of LEA and RED-S. This classification is based on three broad considerations: predominant energy system demands, body weight or physique regulation practices, and training load structure. Using this framework, sports are classified as endurance sports, power and sprint sports, and weight category or aesthetic sports. This categorization is intended to be an interpretive framework rather than a rigid taxonomy. Team sports and composite disciplines may show overlapping features across categories, depending on positional roles, seasonal demands, and body composition pressures.

Endurance Sports

In endurance sports, a prolonged training duration, high training volume, and elevated total energy expenditure create a physiological context in which imbalances between energy intake and expenditure gradually accumulate over time. Therefore, athletes engaged in endurance disciplines are at a particularly high risk of transitioning to LEA, especially when increases in training load are not adequately matched by corresponding increases in energy intake^{25,26}. Importantly, endurance athletes may develop chronic energy deficiency, even in the absence of marked changes in body weight, as weight stability may mask underlying energetic strain^{11,13,27}. A defining feature of LEA in endurance sports is the emergence of physiological and functional disturbances prior to observable weight loss. Experimental studies manipulating EA through short-term energy restriction or increased exercise energy expenditure have demonstrated reductions in luteinizing hormone secretion, thyroid hormone concentrations, and resting metabolic rate despite minimal changes in body mass^{11,16,27,28}. These findings indicate that reduced EA, rather than weight loss per se, alters physiological prioritization. In addition to endocrine regulation, LEA also influences substrate utilization and training adaptation in endurance athletes. Studies conducted under conditions of limited carbohydrate availability have shown impaired muscle glycogen utilization during high-intensity efforts, reduced performance capacity, and delayed recovery, even when the training load is held constant^{25,26}. Over time, sustained LEA has been associated with cumulative health consequences including suppressed bone formation, reduced bone mineral density, and increased stress fracture risk^{11,21,22}. These skeletal risks may be further amplified by the repetitive impact loading inherent to many endurance disciplines^{23,29}.

Collectively, LEA in endurance sports is characterized by gradual energy deficit accumulation with progressive endocrine disruption and performance impairment, reflecting long-term adaptations to insufficient EA. This pattern highlights the importance of a cumulative energy imbalance and chronic metabolic adaptation, closely aligning with the explanatory framework of EBM.

Power and Sprint Sports

LEA in power and sprint sports are frequently characterized by acute functional disturbances rather than pronounced body mass changes. Despite minimal weight fluctuations, athletes in power- and sprint-oriented disciplines remain vulnerable to LEA because of the pressures related to muscle mass maintenance, body composition targets, and inadequate compensatory energy intake^{6,18,30}. Early manifestations of LEA are more likely to be reflected in neuromuscular and performance-related outcomes than in measurable reductions in body weight. Experimental studies involving restricted energy intake or low carbohydrate availability have consistently demonstrated impairments in maximal strength, power output, repeated sprint ability, and post-exercise recovery³¹⁻³³. Importantly, these performance decrements frequently occur in the absence of substantial weight loss, reinforcing the notion that body weight alone is an insensitive marker of energetic strain in both power- and sprint-oriented athletes⁶.

Mechanistically, these functional impairments are likely mediated by limitations in substrate availability and the associated endocrine responses. High-intensity exercise relies heavily on carbohydrate metabolism, and restricted carbohydrate availability has been shown to compromise glycogen utilization during repeated high-intensity efforts, accelerating fatigue onset, and reducing training quality^{25,26}. Beyond simple fuel limitation, reduced carbohydrate availability may also influence neuromuscular function, motor unit recruitment, and central fatigue regulation, thereby amplifying performance impairment³⁴⁻³⁶. From a theoretical perspective, these responses align closely with the mechanistic emphasis on CIM. In power and sprint sports, where performance is tightly coupled with rapid energy turnover and glycolytic flux, disturbances in carbohydrate availability may exert immediate physiological consequences independent of the total energy balance. This framework explains why performance decrements may emerge rapidly even in the absence of measurable body weight changes.

Concurrently, sustained or severe LEA has been associated with a shift toward a more catabolic endocrine environment characterized by reduced anabolic signaling and elevated stress-related hormone levels^{6,20}. Such hormonal alterations may constrain muscle protein synthesis, neuromuscular recovery, and training adaptation, reinforcing the interactions between substrate-mediated limitations and endocrine regulation. Collectively, LEA in power and sprint sports reflects the combined influence of acute substrate availability and systemic regulatory responses rather than only long-term energy imbalance.

Weight-Category and Aesthetic Sports

Athletes competing in weight-category and aesthetic sports operate within environments in which body weight, body fat percentage, and lean mass are integral determinants of competitive success. Consequently, these athletes are often exposed to structurally imposed energy restrictions and intentional body weight manipulations as part of their routine training and competition preparation^{6,37}. Within such

Table 1. Integrative summary of sport-specific expression of LEA and RED-S within the EBM–CIM–EA framework

	Endurance sports	Power/sprint sports	Weight-category & Aesthetic sports
Sport characteristics	High-volume, prolonged training	Short-duration, high-intensity efforts	Performance tied to body mass and physique
Structural origin of LEA	Gradual intake–expenditure mismatch	Inadequate substrate availability	Intentional restriction and repeated weight loss
Dominant model	EBM-dominant	CIM-dominant	Combined EBM-CIM
EBM perspective	Cumulative energy imbalance	Energetic strain despite stable body mass	Sustained deficit from repeated weight manipulation
CIM perspective	Reduced carbohydrate availability	Substrate limits on strength and sprint output	Disturbed partitioning during restriction/refeeding
Metabolic/substrate consequences	Reduced glycogen, delayed recovery	Reduced glycolytic support, faster fatigue	Disruption across restriction-refeeding cycles
Primary physiological responses	Reproductive and thyroid suppression; reduced metabolic rate	Reduced anabolic signaling; impaired neuromuscular recovery	Endocrine suppression; adaptive thermogenesis
Sex-specific considerations	Menstrual and bone-related risks are prominent	Hormonal and recovery effects may differ by sex	Repeated restriction may amplify sex-specific effects
Performance consequences	Reduced endurance performance and adaptation	Reduced strength, power, sprint capacity, and recovery	Reduced explosive performance and agility
RED-S risk profile	Bone stress injury, fatigue, menstrual dysfunction	Lean mass loss, impaired recovery	Hormonal disruption, bone injury risk, weight cycling
Role of EA	Links long-term energy balance with substrate use	Determines substrate availability for high-intensity output	Balances restriction and physiological recovery

LEA: low energy availability, EBM: energy balance model, CIM: Carbohydrate–insulin model, RED-S: relative energy deficiency in sport, EA: energy availability.

contexts, the risk of chronic LEA is particularly pronounced and independent of overall training volume²⁷.

Rapid weight loss (RWL) strategies, commonly employed to meet weight or aesthetic criteria, typically involve combinations of energy restriction, dehydration, fluid manipulation, and intensified training loads. Empirical investigations of these practices have documented concurrent reductions in EA, disturbances in body fluid balance, and alterations in key endocrine markers, including testosterone, thyroid hormones, and leptin^{33,38,39}. These physiological perturbations are closely associated with impaired recovery, and decreased training and competition performance, indicating that the biological cost of weight manipulation extends beyond transient changes in body mass. Repeated cycles of energy restriction and weight gain further exacerbate the metabolic and health risks. Longitudinal studies have shown that post-restriction recovery is characterized by adaptive thermogenesis and persistent alterations in appetite and energy partitioning, leading to preferential fat regaining relative to lean mass, a phenomenon often described as overshoot⁴⁰. This pattern increases the pressure for subsequent weight-reduction cycles and contributes to the cumulative deterioration of body composition. These findings illustrate the concurrent operation of long-term energy-conserving adaptations and short-term substrate–hormonal disruptions, reflecting the combined influence of EBM- and CIM-related mechanisms.

The performance consequences of RWL are also substantial. Reductions in sprint capacity, reaction time, agility,

rate of force development, and jump performance have been reported following RWL, along with impairments in decision-making and tactical responsiveness during competitions^{33,41}. These results suggest that the limitations in substrate availability, neuromuscular function, and recovery outweigh any transient competitive advantage conferred by reduced body mass.

In summary, weight category and aesthetic sports represent contexts in which chronic LEA are structurally embedded within training and competitive practices. In these environments, long-term energy deficiency and short-term disruptions in substrate utilization and hormonal regulation accumulate concurrently, resulting in pronounced amplification of RED-S risk. To facilitate the following integrative discussion, Table 1 summarizes the sports-specific expressions of LEA and RED-S within the proposed EBM–CIM–EA framework.

DISCUSSION

As summarized in Table 1, the sport-specific manifestations of LEA and RED-S support the view that these conditions represent integrated physiological phenomena that cannot be adequately explained by a single model or weight-centric perspective. In athletic populations, the long-term regulation of energy balance and short-term modulation of substrate utilization operate concurrently, with EA functioning as the central regulatory variable. This integrat-

ed interpretation provides a more physiologically coherent account of the disturbances observed under LEA, including endocrine suppression, altered substrate utilization, and impaired performance capacity.

Within this framework, EBM and CIM are best interpreted as complementary explanatory domains rather than as competing mechanisms. EBM offers a basis for understanding cumulative regulatory adaptations arising from sustained energetic strain, whereas CIM provides mechanistic insights into the acute metabolic responses associated with substrate partitioning and hormonal signaling. Therefore, the physiological consequences of LEA are more consistently understood when long-term regulatory processes and short-term metabolic dynamics are jointly considered. EA provides a unifying construct that links these processes, and represents the interface at which chronic energy regulation converges with acute metabolic modulation.

Sport-specific observations further reinforce this integrative interpretation. The heterogeneous manifestations of LEA across endurance, power/sprint, and weight-category/aesthetic disciplines indicate that identical reductions in EA may produce distinct patterns of physiological and performance disruption depending on metabolic demands and training structure. Endurance sports primarily reflect cumulative regulatory adaptations associated with sustained energetic strain, whereas power and sprint sports demonstrate performance disturbances that are closely tied to substrate-mediated limitations. In turn, weight category and aesthetic sports exhibit compounded disruptions arising from chronic restrictions and cyclical metabolic perturbations. Collectively, these patterns support the interpretation of LEA as a systemic regulatory state, rather than a singular metabolic or nutritional phenomenon.

This framework also helps to position LEA and RED-S in relation to earlier and more recent conceptual models. The female athlete triad provides an important foundation for identifying the interrelationships among energy deficiency, menstrual dysfunction, and impaired bone health²². The 2023 IOC RED-S consensus statement expanded this perspective by recognizing that LEA can disrupt multiple physiological systems in both female and male athletes⁵. The present EA-centered framework is consistent with this broader view. This further suggests that these outcomes can be interpreted more coherently when long-term regulatory adaptation and short-term metabolic dynamics are considered together, rather than separately.

This interpretation may also help reconcile apparent inconsistencies in the literature. Some studies have emphasized the importance of chronic energetic strain, adaptive thermogenesis, and cumulative endocrine suppression. Other studies have focused on the acute changes in carbohydrate availability, substrate partitioning, and performance. However, these perspectives are not necessarily contradictory. Instead, they reflect the different temporal and physiological levels of the analysis. Within this framework, EBM primarily explains chronic regulatory adaptation, whereas CIM explains short-term metabolic dynamics. EA functions as an integrative interface that links these domains.

From a physiological standpoint, the present framework emphasizes that performance regulation, metabolic adaptation, and health outcomes arise from dynamic interactions rather than from singular mechanisms. Athletes may exhibit substantial disturbances in endocrine function and exercise capacity even when body mass remains stable. This pattern highlights the limitations of weight-centric interpretation. Therefore, LEA should not be viewed solely as an energy intake problem or a body composition issue, but as a systemic regulatory state emerging from sustained mismatches between energetic demand and availability.

From an applied perspective, this framework highlights the limitations of approaches that primarily rely on body weight and short-term dietary intake. Athletes may show meaningful endocrine, metabolic, and performance disturbances even when body mass remains stable. Therefore, monitoring strategies should consider EA-related indicators, along with training load, recovery status, and sport-specific performance demands. Nutritional support should also be aligned with the metabolic characteristics of each sport and fluctuations in training volume and intensity.

This study has several limitations. The available literature does not fully capture the diversity of sports disciplines, populations, or methodological approaches, and not all studies have directly quantified EA. In addition, the three-category classification adopted in this review does not fully capture LEA risk profiles specific to team sports or composite-discipline athletes, whose sport-specific manifestations may require dedicated examination in future research. Nevertheless, the consistency of the findings across the experimental and applied contexts supports the value of an EA-centered interpretation. There are several directions for future research that follow this framework. First, sex- and sport-specific EA thresholds should be examined more carefully across athletic disciplines. Second, more work is needed to understand how training periodization interacts with EA fluctuations. Third, longitudinal studies should investigate the transition from acute to chronic LEA and RED-S. Fourth, field-applicable biomarkers and practical monitoring tools are required to assess EA in real time.

In conclusion, LEA and RED-S can be most coherently interpreted as integrated physiological expressions of disrupted EA. Rather than representing isolated metabolic or hormonal disturbances, these conditions reflect systemic regulatory adaptations shaped by temporal and metabolic constraints. The EA-centered integrative model proposed herein provides a unified theoretical basis for interpreting metabolic adaptation, performance regulation, and health outcomes in athletes.

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